Vortrix Physics Quick Manual – Navigating the Application

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Minimum Requirements

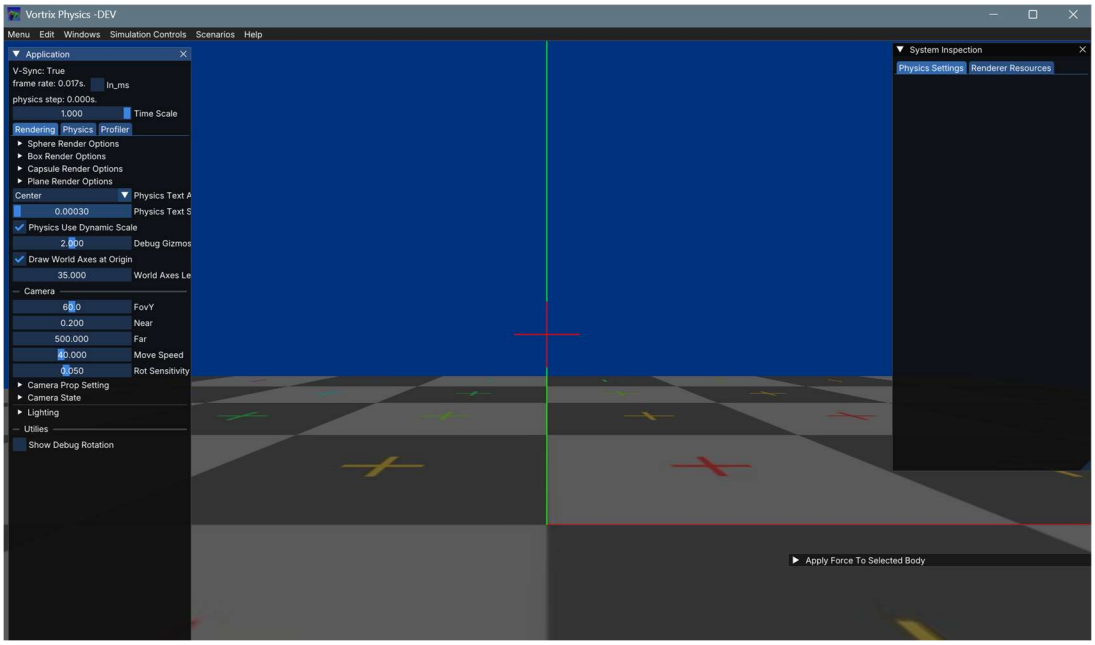
- Windows x64.
- Processor (CPU): Intel 4th Generation Haswell / AMD Zen or newer (AVX2 and FMA support).
- Graphics (GPU): OpenGL hardware.

If custom Installing/Compiling through source code, see **Vortrix Physics Quick Manual – Installation (Project Generation and Compilation)** for detailed minimum requirements.

Launching the Application

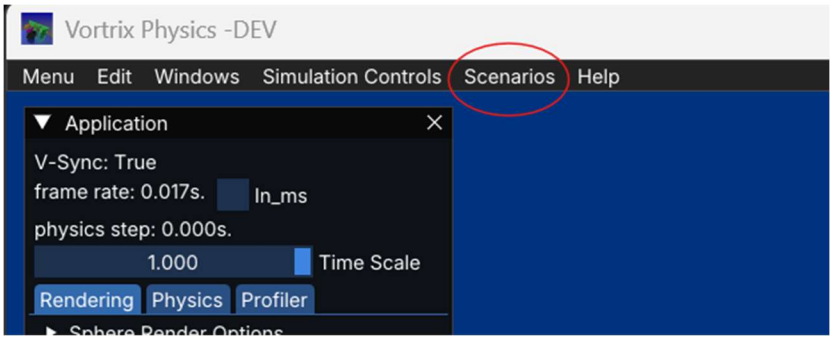
Run the executable provided, “VortrixPhysicsSample.exe”, or build from the source code. See Vortrix Physics Quick Manual – Installation.

This launches the physics viewport.

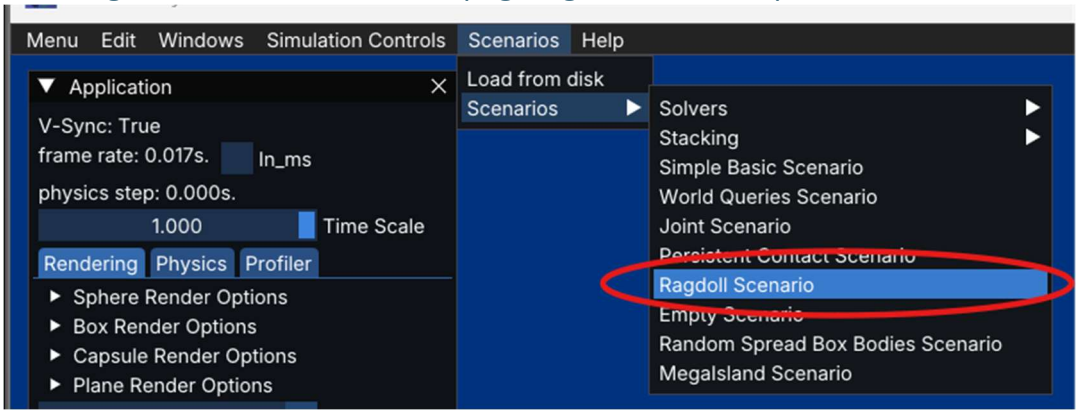


Loading Scenarios

Scenarios within the Vortrix Physics Engine are split into **Pre-coded internal simulations** and **saved scenarios loaded from disk (.json)**.

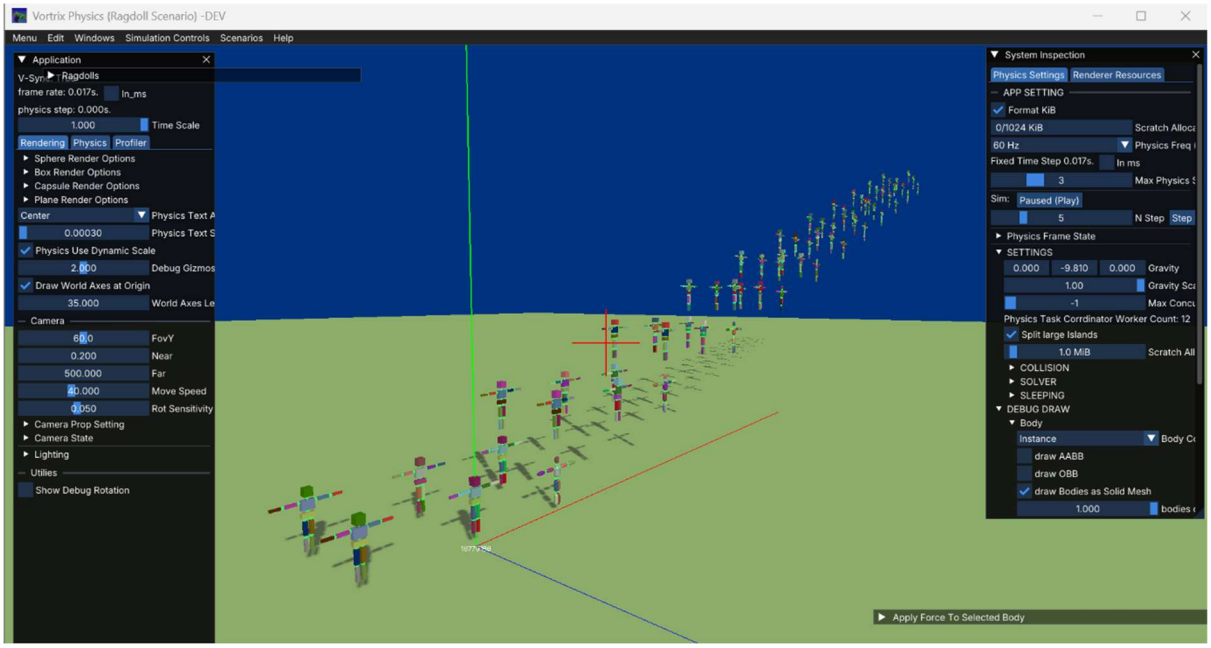


Loading Pre-Coded Scenarios (e.g Ragdoll Scenario)

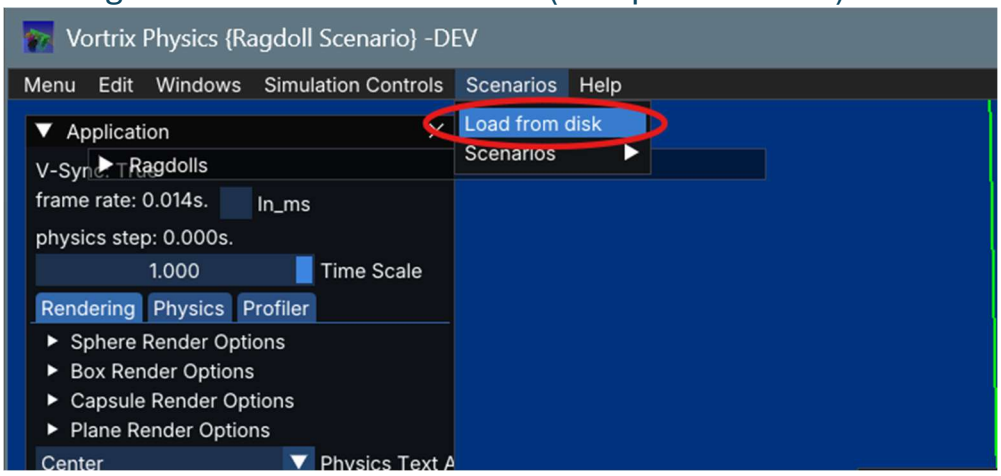


To load the built-in Ragdoll testing scenario:

- Click **Scenarios** on the top menu bar.
- Hover over nested **Scenarios** dropdown item.
- Select **Ragdoll Scenario** from the list
- (Scenarios -> Scenarios -> Ragdoll Scenario)



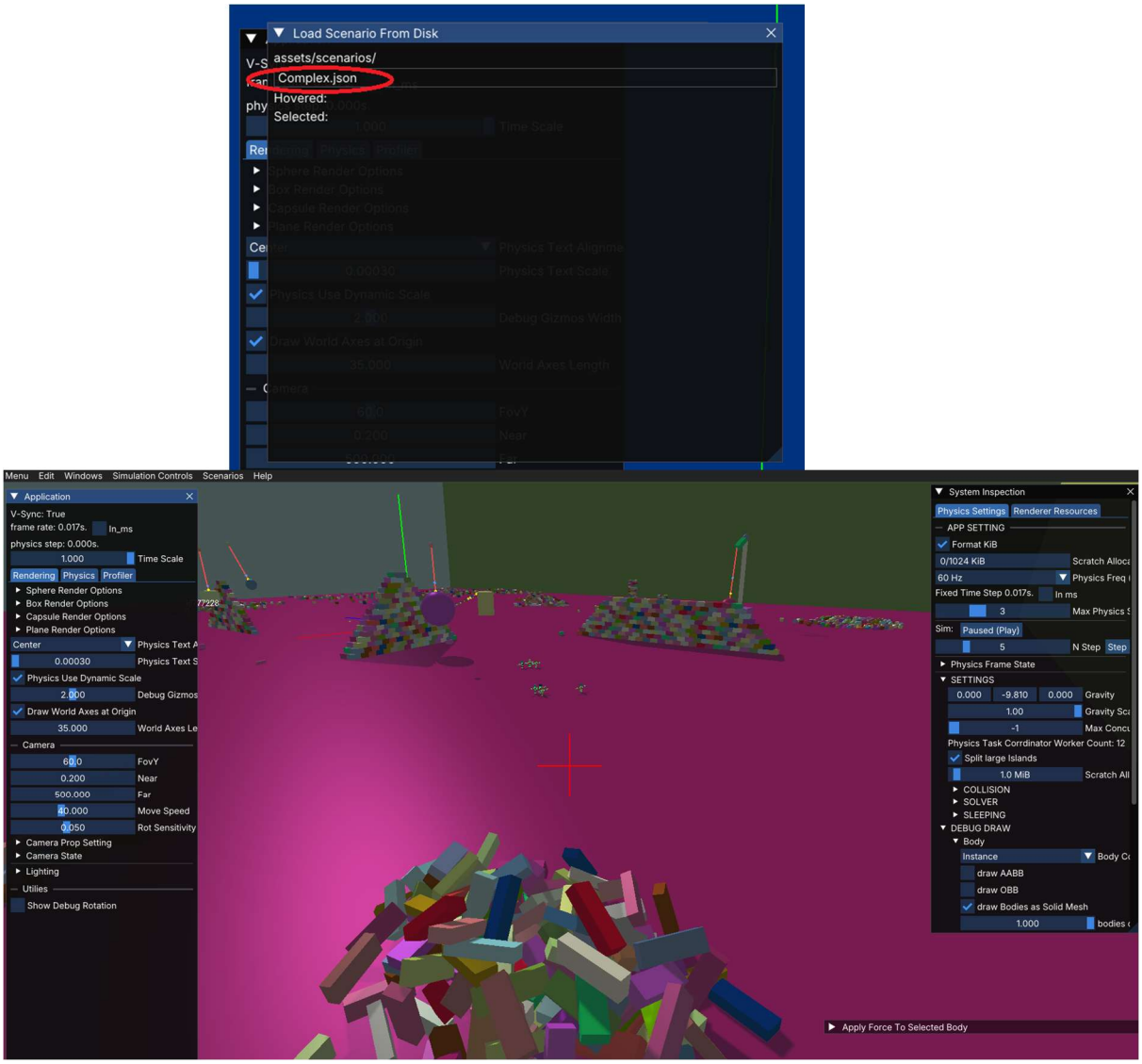
Loading Saved Scenarios From Disk (Complex Scenario)



To load external JSON saved scenarios from file

- Click **Scenarios** on the top menu bar.
- Click **Load from disk** from the drop-down menu
- In the pop-up utility window, double-click **Complex.json**

- Close the **Load Scenario From Disk** pop-up window.



Application Controls & Camera Navigation

Use these inputs to inspect the active physics viewport and control engine execution:

- **Mouse Right-Click**: Toggles the mouse cursor lock status (Unlocks for UI interaction/Locks for camera look).
- **P / PAUSE Key:** Pauses or resumes the active physics simulation.
- **W / A / S / D:** Translates the viewport camera **Forward / Left / Backward / Right** relative to its local frame.
- **Q / E:** Translates the viewport camera **Downward / Upward.**
- **Ctrl + R:** Flushes and reloads the currently active simulation scenario back to its initial state.

Diagnostic Hotkeys & Help

- **Ctrl + V:** Toggles Vertical Synchronisation (V-Sync) on or off to test uncapped frame rates.
- **F10:** Advances the simulation by exactly one single discrete frame step.

- **Help menu:** Located on the top menu bar for further engine feature manual details.

